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JEE MAIN · QUESTION PAPER

JEE Main 2023 (Online) 24th January Morning Shift

JEE Main Question Paper with Answer Key

24 Jan 2023	90	20	English
HELD ON	QUESTIONS	OUT OF SYLLABUS	LANGUAGE

INSIDE THIS PDF

Chemistry	23 questions
Mathematics	21 questions
Physics	26 questions
Out of syllabus	20 questions
Answer key	at the end

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Chemistry 23 questions

Q1. MCQ (Single Correct Answer)

The primary and secondary valencies of cobalt respectively in $[Co(NH_3)_5Cl]Cl_2$ are :

(A) 2 and 8 (B) 2 and 6 (C) 3 and 5 (D) 3 and 6

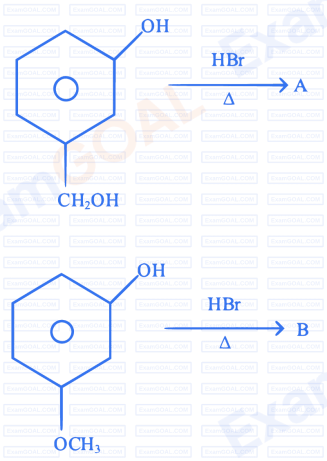
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Q2. MCQ (Single Correct Answer)

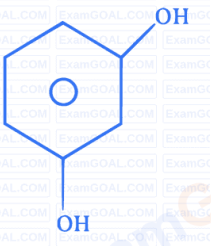
Open solution

'A' and 'B' formed in the following set of reactions are :



(A) A =  , B = 

(B) A =  , B = 

(C) A =  , B = 

(D) A =  , B = 

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Q3. MCQ (Single Correct Answer)

Open solution

Decreasing order of the hydrogen bonding in following forms of water is correctly represented by

A. Liquid water

B. Ice

C. Impure water

Choose the correct answer from the options given below :

(A) A = B > C **(B)** B > A > C **(C)** A > B > C **(D)** C > B > A

Q4. MCQ (Single Correct Answer)

Open solution

Assertion A : Hydrolysis of an alkyl chloride is a slow reaction but in the presence of NaI, the rate of the hydrolysis increases.

Reason R : I^- is a good nucleophile as well as a good leaving group.

In the light of the above statements, choose the correct answer from the options given below

(A) A is true but R is false

(B) A is false but R is true

(C) Both A and R are true and R is the correct explanation of A

(D) Both A and R are true but R is NOT the correct explanation of A

Q5. MCQ (Single Correct Answer)

Open solution

An ammoniacal metal salt solution gives a brilliant red precipitate on addition of dimethylglyoxime. The metal ion is :

(A) Co^{2+} **(B)** Fe^{2+} **(C)** Ni^{2+} **(D)** Cu^{2+}

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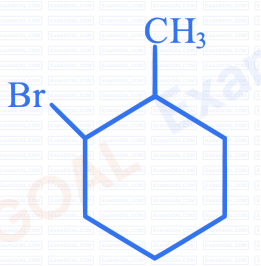
Q6. MCQ (Single Correct Answer)

Open solution

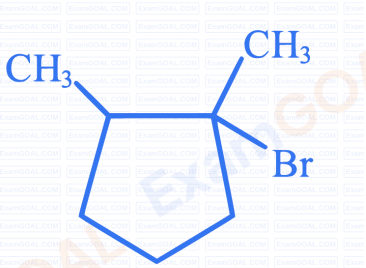
In the following given reaction, 'A' is



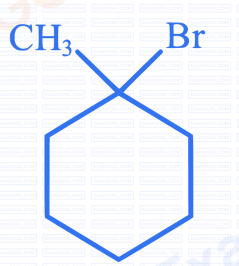
(A)



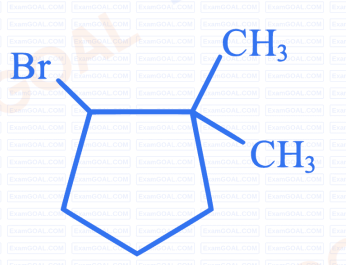
(B)



(C)



(D)

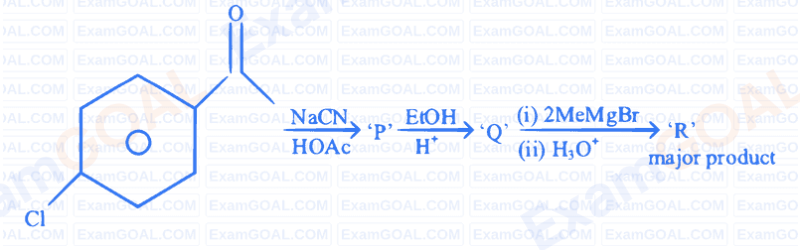


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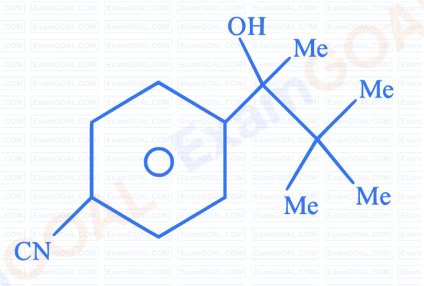
Q7. MCQ (Single Correct Answer)

Open solution

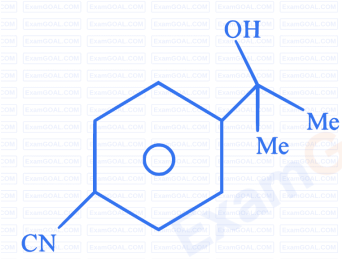
'R' formed in the following sequence of reactions is :



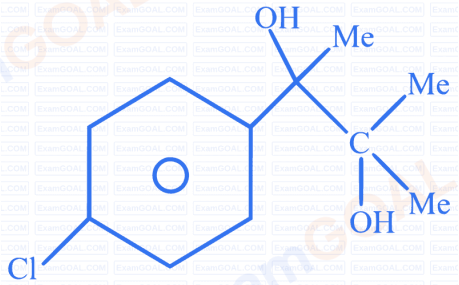
(A)



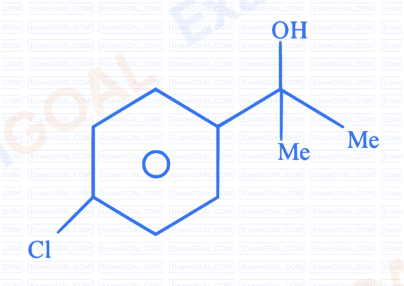
(B)



(C)



(D)



Q8. MCQ (Single Correct Answer)

Open solution

The magnetic moment of a transition metal compound has been calculated to be 3.87 B.M. The metal ion is

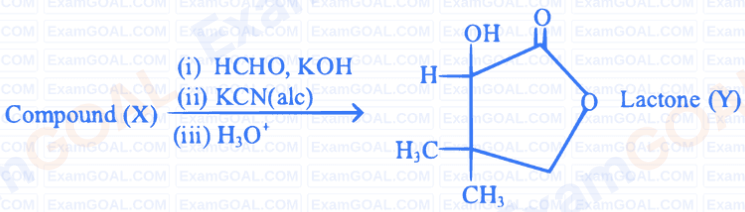
(A) Ti^{2+} (B) V^{2+} (C) Cr^{2+} (D) Mn^{2+}

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Q9. MCQ (Single Correct Answer)

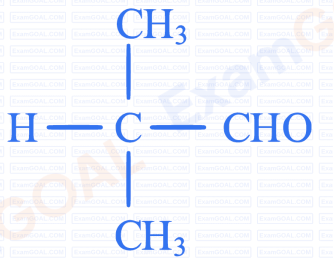
Open solution

Compound (X) undergoes following sequence of reactions to give the Lactone (Y).



Compound (X) is :

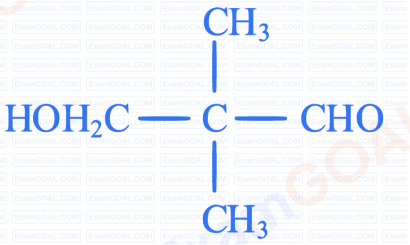
(A)



(B)



(C)



(D)



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Q10. MCQ (Single Correct Answer)

Open solution

In the depression of freezing point experiment

A. Vapour pressure of the solution is less than that of pure solvent

B. Vapour pressure of the solution is more than that of pure solvent

C. Only solute molecules solidify at the freezing point

D. Only solvent molecules solidify at the freezing point

Choose the most appropriate answer from the options given below :

(A) A and C only **(B)** A only **(C)** A and D only **(D)** B and C only

Q11. MCQ (Single Correct Answer)

Open solution

Order of Covalent bond;

A. $\mathrm{KF} > \mathrm{KI}; \mathrm{LiF} > \mathrm{KF}$

B. $\mathrm{KF} < \mathrm{KI}; \mathrm{LiF} > \mathrm{KF}$

C. $\mathrm{SnCl_4} > \mathrm{SnCl_2}; \mathrm{CuCl} > \mathrm{NaCl}$

D. $\mathrm{LiF} > \mathrm{KF}; \mathrm{CuCl} < \mathrm{NaCl}$

E. $\mathrm{KF} < \mathrm{KI}; \mathrm{CuCl} > \mathrm{NaCl}$

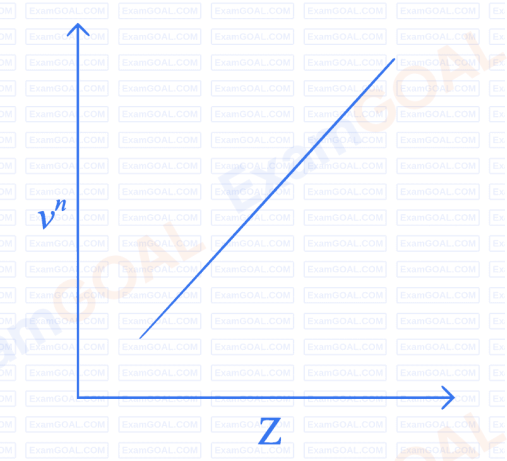
Choose the correct answer from the options given below :

(A) B, C only **(B)** A, B only **(C)** C, E only **(D)** B, C, E only

Q12. MCQ (Single Correct Answer)

Open solution

It is observed that characteristic X-ray spectra of elements show regularity. When frequency to the power "n" i.e. v^n of X-rays emitted is plotted against atomic number "Z", following graph is obtained.



The value of 'n' is :

(A) 1 **(B)** $\frac{1}{2}$ **(C)** 2 **(D)** 3

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Q13. MCQ (Single Correct Answer)

Open solution

Increasing order of stability of the resonance structures is :

A. 

B. 

C. 

D. 

Choose the correct answer from the options given below :

(A) D, C, B, A **(B)** C, D, B, A **(C)** C, A, B, D **(D)** D, C, A, B

Q14. Numerical

Open solution

For independent processes at 300 K

Process	$\Delta H/kJ\ mol^{-1}$	$\Delta S/J\ K^{-1}$
A	-25	-80
B	-22	40
C	25	-50
D	22	20

The number of non-spontaneous processes from the following is _____

Q15. Numerical

Open solution

When $Fe_{0.93}O$ is heated in presence of oxygen, it converts to Fe_2O_3. The number of correct statement/s from the following is _____

A. The equivalent weight of $Fe_{0.93}O$ is $\frac{\text{Molecular weight}}{0.79}$

B. The number of moles of Fe^{2+} and Fe^{3+} in 1 mole of $Fe_{0.93}O$ is 0.79 and 0.14 respectively

C. $Fe_{0.93}O$ is metal deficient with lattice comprising of cubic closed packed arrangement of O^{2-} ions

D. The % composition of Fe^{2+} and Fe^{3+} in $Fe_{0.93}O$ is 85% and 15% respectively

Q16. Numerical

Open solution

5 g of NaOH was dissolved in deionized water to prepare a 450 mL stock solution. What volume (in mL) of this solution would be required to prepare 500 mL of 0.1 M solution? _____

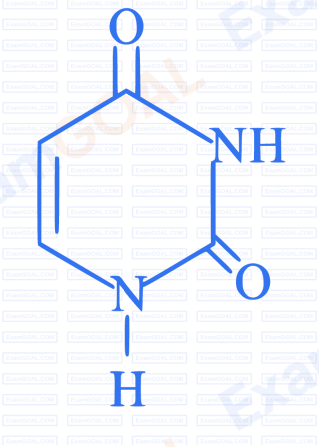
Given : Molar Mass of Na, O and H is 23, 16 and 1 g mol^{-1} respectively

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Q17. Numerical

Uracil is a base present in RNA with the following structure. % of N in uracil is _____

Open solution



Given:

Molar mass N = 14 g mol^{-1}

O = 16 g mol^{-1}

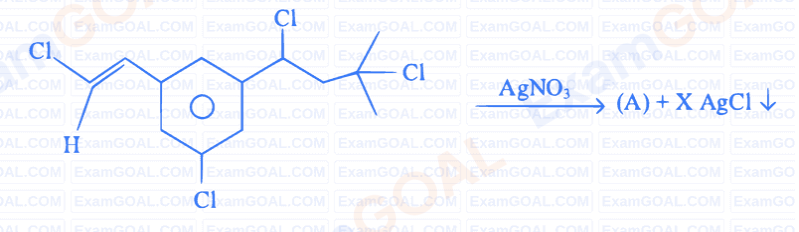
C = 12 g mol^{-1}

H = 1 g mol^{-1}

Q18. Numerical

Number of moles of AgCl formed in the following reaction is _____

Open solution



Q19. Numerical

The number of correct statement/s from the following is _____

Open solution

A. Larger the activation energy, smaller is the value of the rate constant.

B. The higher is the activation energy, higher is the value of the temperature coefficient.

C. At lower temperatures, increase in temperature causes more change in the value of k than at higher temperature

D. A plot of $\ln k$ vs $\frac{1}{T}$ is a straight line with slope equal to $-\frac{E_a}{R}$

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Q20. Numerical

The dissociation constant of acetic acid is $x \times 10^{-5}$. When 25 mL of 0.2 M CH_3COONa solution is mixed with 25 mL of 0.02 M CH_3COOH solution, the pH of the resultant solution is found to be equal to 5. The value of x is _____

Open solution

Q21. Numerical

At 298 K, a 1 litre solution containing 10 mmol of $Cr_2O_7^{2-}$ and 100 mmol of Cr^{3+} shows a pH of 3.0.

Given : $Cr_2O_7^{2-} \rightarrow Cr^{3+}$; $E^\circ = 1.330V$

and $\frac{2.303\,RT}{F} = 0.059$ V

The potential for the half cell reaction is $x \times 10^{-3}$ V. The value of x is _____

Open solution

Q22. Numerical

The d-electronic configuration of $[CoCl_4]^{2-}$ in tetrahedral crystal field is $e^m t_2^n$. Sum of "m" and "number of unpaired electrons" is _____

Open solution

Q23. Numerical

If wavelength of the first line of the Paschen series of hydrogen atom is 720 nm, then the wavelength of the second line of this series is _____ nm. (Nearest integer)

Open solution

Mathematics 21 questions

Q24. MCQ (Single Correct Answer)

Let $p, q \in \mathbb{R}$ and $\left(1-\sqrt{3}i\right)^{200} = 2^{199}(p+iq), i = \sqrt{-1}$ then $p+q+q^2$ and $p-q+q^2$ are roots of the equation.

(A) $x^2+4x-1=0$ (B) $x^2-4x+1=0$ (C) $x^2+4x+1=0$ (D) $x^2-4x-1=0$

Open solution

Q25. MCQ (Single Correct Answer)

Let N denote the number that turns up when a fair die is rolled. If the probability that the system of equations

$x+y+z=1$

$2x+Ny+2z=2$

$3x+3y+Nz=3$

has unique solution is $\frac{k}{6}$, then the sum of value of k and all possible values of N is :

(A) 18 (B) 21 (C) 20 (D) 19

Open solution

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Q26. MCQ (Single Correct Answer)

The relation $R = \{(a, b) : \gcd(a, b) = 1, 2a \neq b, a, b \in \mathbb{Z}\}$ is :

(A) reflexive but not symmetric

(B) transitive but not reflexive

(C) symmetric but not transitive

(D) neither symmetric nor transitive

Open solution

Q27. MCQ (Single Correct Answer)

$\lim\limits_{t\to 0}\left(1^{\frac{1}{\sin^2 t}} + 2^{\frac{1}{\sin^2 t}} + \ldots + n^{\frac{1}{\sin^2 t}}\right)^{\sin^2 t}$ is equal to

(A) $\frac{n(n+1)}{2}$

(B) n

(C) $n^2 + n$

(D) n^2

Open solution

Q28. MCQ (Single Correct Answer)

$\tan^{-1}\left(\frac{1+\sqrt{3}}{3+\sqrt{3}}\right) + \sec^{-1}\left(\sqrt{\frac{8+4\sqrt{3}}{6+3\sqrt{3}}}\right)$ is equal to :

(A) $\frac{\pi}{2}$

(B) $\frac{\pi}{3}$

(C) $\frac{\pi}{6}$

(D) $\frac{\pi}{4}$

Open solution

Q29. MCQ (Single Correct Answer)

Let $y = y(x)$ be the solution of the differential equation $x^3dy + (xy - 1)dx = 0, x > 0, y\left(\frac{1}{2}\right) = 3 - e$. Then y (1) is equal to

(A) $2 - e$

(B) 3

(C) 1

(D) e

Open solution

Q30. MCQ (Single Correct Answer)

Let Ω be the sample space and $A \subseteq \Omega$ be an event.

Given below are two statements :

(S1) : If P(A) = 0, then A = ϕ

(S2) : If P(A) = 1, then A = Ω

Then :

(A) both (S1) and (S2) are true

(B) both (S1) and (S2) are false

(C) only (S2) is true

(D) only (S1) is true

Open solution

Q31. MCQ (Single Correct Answer)

The area enclosed by the curves $y^2 + 4x = 4$ and $y - 2x = 2$ is :

(A) $\frac{22}{3}$

(B) 9

(C) $\frac{23}{3}$

(D) $\frac{25}{3}$

Open solution

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Q32. MCQ (Single Correct Answer)

The equation $x^2 - 4x + [x] + 3 = x[x]$, where $[x]$ denotes the greatest integer function, has :

(A) exactly two solutions in $(-\infty, \infty)$

(B) no solution

(C) a unique solution in $(-\infty, \infty)$

(D) a unique solution in $(-\infty, 1)$

Open solution

Q33. MCQ (Single Correct Answer)

If A and B are two non-zero n $\times$ n matrices such that $A^2 + B = A^2B$, then :

(A) $A^2B = I$

(B) $A^2 = I$ or $B = I$

(C) $A^2B = BA^2$

(D) $AB = I$

Open solution

Q34. MCQ (Single Correct Answer)

For three positive integers p, q, r, $x^{pq^2} = y^{qr} = z^{p^2r}$ and r = pq + 1 such that 3, 3 $\log_y x$, 3 $\log_z y$, 7 $\log_x z$ are in A.P. with common difference $\frac{1}{2}$. Then r-p-q is equal to

(A) 12

(B) -6

(C) 6

(D) 2

Open solution

Q35. MCQ (Single Correct Answer)

Let PQR be a triangle. The points A, B and C are on the sides QR, RP and PQ respectively such that $\frac{QA}{AR} = \frac{RB}{BP} = \frac{PC}{CQ} = \frac{1}{2}$. Then $\frac{Area(\Delta PQR)}{Area(\Delta ABC)}$ is equal to :

(A) $\frac{5}{2}$

(B) 4

(C) 2

(D) 3

Open solution

Q36. MCQ (Single Correct Answer)

Let $f(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right) & , x \neq 0 \\ 0 & , x = 0 \end{cases}$

Then at $x = 0$

(A) f is continuous but f' is not continuous

(B) f and f' both are continuous

(C) f is continuous but not differentiable

(D) f' is continuous but not differentiable

Open solution

Q37. Numerical

Let $\lambda \in \mathbb{R}$ and let the equation E be $|x|^2 - 2|x| + |\lambda - 3| = 0$. Then the largest element in the set S = $\{x + \lambda : x$ is an integer solution of E$\}$ is _____

Open solution

Q38. Numerical

The shortest distance between the lines $\frac{x-2}{3} = \frac{y+1}{2} = \frac{z-6}{2}$ and $\frac{x-6}{3} = \frac{1-y}{2} = \frac{z+8}{0}$ is equal to _____

Open solution

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Q39. Numerical

The 4th term of GP is 500 and its common ratio is $\frac{1}{m}, m \in \mathbb{N}$. Let S_n denote the sum of the first n terms of this GP. If $S_6 > S_5 + 1$ and $S_7 < S_6 + \frac{1}{2}$, then the number of possible values of m is _____

Open solution

Q40. Numerical

Let C be the largest circle centred at (2, 0) and inscribed in the ellipse $\frac{x^2}{36} + \frac{y^2}{16} = 1$. If $(1, \alpha)$ lies on C, then $10\,\alpha^2$ is equal to _____

Open solution

Q41. Numerical

A boy needs to select five courses from 12 available courses, out of which 5 courses are language courses. If he can choose at most two language courses, then the number of ways he can choose five courses is _____

Open solution

Q42. Numerical

The value of $12\int_0^3 |x^2 - 3x + 2|dx$ is _____

Open solution

Q43. Numerical

The value of $\frac{8}{\pi}\int_0^{\frac{\pi}{2}} \frac{(\cos x)^{2023}}{(\sin x)^{2023}+(\cos x)^{2023}}dx$ is _____

Open solution

Q44. Numerical

The number of 9 digit numbers, that can be formed using all the digits of the number 123412341 so that the even digits occupy only even places, is _____.

Open solution

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Physics 26 questions

Q45. MCQ (Single Correct Answer)

Open solution

Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R

Assertion A : Photodiodes are preferably operated in reverse bias condition for light intensity measurement.

Reason R : The current in the forward bias is more than the current in the reverse bias for a $p-n$ junction diode.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both A and R are true and R is the correct explanation of A

(B) A is false but R is true

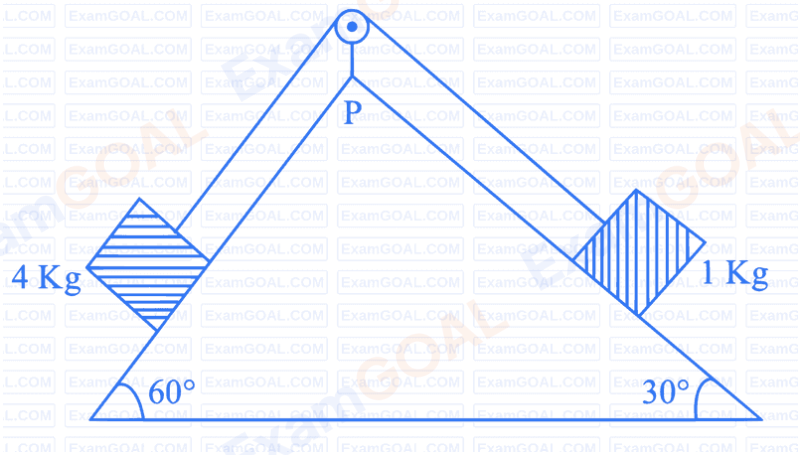
(C) A is true but R is false

(D) Both A and R are true but R is NOT the correct explanation of A

Q46. MCQ (Single Correct Answer)

Open solution

As per given figure, a weightless pulley P is attached on a double inclined frictionless surfaces. The tension in the string (massless) will be (if g = 10 m/s^2)



(A) $\left(4\sqrt{3}-1\right)N$ (B) $\left(4\sqrt{3}+1\right)N$ (C) $4\left(\sqrt{3}-1\right)N$ (D) $4\left(\sqrt{3}+1\right)N$

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Q47. MCQ (Single Correct Answer)

Open solution

Given below are two statements :

Statement I : If the Brewster's angle for the light propagating from air to glass is θ_{B}, then the Brewster's angle for the light propagating from glass to air is $\frac{\pi}{2} - \theta_B$

Statement II : The Brewster's angle for the light propagating from glass to air is $\tan^{-1}(\mu_{\mathrm{g}})$ where μ_{g} is the refractive index of glass.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are false

(B) Both Statement I and Statement II are true

(C) Statement I is false but Statement II is true

(D) Statement I is true but Statement II is false

Q48. MCQ (Single Correct Answer)

Open solution

A 100 m long wire having cross-sectional area $6.25 \times 10^{-4}\ \mathrm{m}^2$ and Young's modulus is $10^{10}\ \mathrm{Nm}^{-2}$ is subjected to a load of 250 N, then the elongation in the wire will be :

(A) $6.25 \times 10^{-6}\ \mathrm{m}$ (B) $4 \times 10^{-3}\ \mathrm{m}$ (C) $4 \times 10^{-4}\ \mathrm{m}$ (D) $6.25 \times 10^{-3}\ \mathrm{m}$

Q49. MCQ (Single Correct Answer)

Open solution

The weight of a body at the surface of earth is 18 N. The weight of the body at an altitude of 3200 km above the earth's surface is (given, radius of earth $\mathrm{R_e} = 6400\ \mathrm{km}$) :

(A) 9.8 N (B) 4.9 N (C) 19.6 N (D) 8 N

Q50. MCQ (Single Correct Answer)

Open solution

From the photoelectric effect experiment, following observations are made. Identify which of these are correct.

A. The stopping potential depends only on the work function of the metal.

B. The saturation current increases as the intensity of incident light increases.

C. The maximum kinetic energy of a photo electron depends on the intensity of the incident light.

D. Photoelectric effect can be explained using wave theory of light.

Choose the correct answer from the options given below :

(A) A, B, D only (B) A, C, D only (C) B, C only (D) B only

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Q51. MCQ (Single Correct Answer)

Two long straight wires P and Q carrying equal current 10A each were kept parallel to each other at 5 cm distance. Magnitude of magnetic force experienced by 10 cm length of wire P is F_1. If distance between wires is halved and currents on them are doubled, force F_2 on 10 cm length of wire P will be:

Open solution

(A) $\frac{F_1}{8}$ (B) 10 F_1 (C) $\frac{F_1}{10}$ (D) 8 F_1

Q52. MCQ (Single Correct Answer)

A circular loop of radius r is carrying current I A. The ratio of magnetic field at the center of circular loop and at a distance r from the center of the loop on its axis is :

Open solution

(A) $3\sqrt{2}:2$ (B) $1:3\sqrt{2}$ (C) $2\sqrt{2}:1$ (D) $1:\sqrt{2}$

Q53. MCQ (Single Correct Answer)

If two charges q_1 and q_2 are separated with distance 'd' and placed in a medium of dielectric constant K. What will be the equivalent distance between charges in air for the same electrostatic force?

Open solution

(A) $d\sqrt{k}$ (B) $1.5d\sqrt{k}$ (C) $k\sqrt{d}$ (D) $2d\sqrt{k}$

Q54. MCQ (Single Correct Answer)

A conducting circular loop of radius $\frac{10}{\sqrt{\pi}}$ cm is placed perpendicular to a uniform magnetic field of 0.5 T. The magnetic field is decreased to zero in 0.5 s at a steady rate. The induced emf in the circular loop at 0.25 s is :

Open solution

(A) emf = 10 mV (B) emf = 5 mV (C) emf = 100 mV (D) emf = 1 mV

Q55. MCQ (Single Correct Answer)

1 g of a liquid is converted to vapour at 3×10^5 Pa pressure. If 10% of the heat supplied is used for increasing the volume by 1600 cm^3 during this phase change, then the increase in internal energy in the process will be :

Open solution

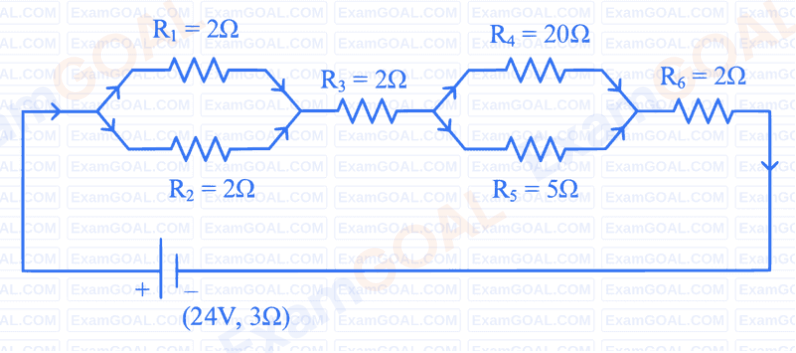
(A) 4800 J (B) 4320 J (C) 432000 J (D) 4.32×10^8 J

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Q56. MCQ (Single Correct Answer)

Open solution

As shown in the figure, a network of resistors is connected to a battery of 24V with an internal resistance of 3 Ω. The currents through the resistors R_4 and R_5 are I_4 and I_5 respectively. The values of I_4 and I_5 are :



(A) $I_4 = \frac{8}{5}A$ and $I_5 = \frac{2}{5}A$

(B) $I_4 = \frac{6}{5}A$ and $I_5 = \frac{24}{5}A$

(C) $I_4 = \frac{2}{5}A$ and $I_5 = \frac{8}{5}A$

(D) $I_4 = \frac{24}{5}A$ and $I_5 = \frac{6}{5}A$

Q57. MCQ (Single Correct Answer)

Open solution

A travelling wave is described by the equation

$y(x,t) = [0.05\sin(8x - 4t)]$ m

The velocity of the wave is : [all the quantities are in SI unit]

(A) $4\ ms^{-1}$ (B) $2\ ms^{-1}$ (C) $8\ ms^{-1}$ (D) $0.5\ ms^{-1}$

Q58. MCQ (Single Correct Answer)

Open solution

The maximum vertical height to which a man can throw a ball is 136 m. The maximum horizontal distance upto which he can throw the same ball is :

(A) 136 m (B) 272 m (C) 68 m (D) 192 m

Q59. MCQ (Single Correct Answer)

Open solution

Given below are two statements :

Statement I : The temperature of a gas is -73°C. When the gas is heated to 527°C, the root mean square speed of the molecules is doubled.

Statement II : The product of pressure and volume of an ideal gas will be equal to translational kinetic energy of the molecules.

In the light of the above statements, choose the correct answer from the options given below :

(A) Statement I is true but Statement II is false

(B) Both Statement I and Statement II are true

(C) Statement I is false but Statement II is true

(D) Both Statement I and Statement II are false

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Q60. MCQ (Single Correct Answer)

Open solution

Match List I with List II

	List-I		List-II
A.	Planck's constant (h)	I.	$[\mathrm{M}^1\,\mathrm{L}^2\,\mathrm{T}^{-2}]$
B.	Stopping potential (Vs)	II.	$[\mathrm{M}^1\,\mathrm{L}^1\,\mathrm{T}^{-1}]$
C.	Work function (ϕ)	III.	$[\mathrm{M}^1\,\mathrm{L}^2\,\mathrm{T}^{-1}]$
D.	Momentum (p)	IV.	$[\mathrm{M}^1\,\mathrm{L}^2\,\mathrm{T}^{-3}\,\mathrm{A}^{-1}]$

Choose the correct answer from the options given below :

(A) A-I, B-III, C-IV, D-II

(B) A-III, B-IV, C-I, D-II

(C) A-II, B-IV, C-III, D-I

(D) A-III, B-I, C-II, D-IV

Q61. MCQ (Single Correct Answer)

Open solution

Given below are two statements :

Statement I : An elevator can go up or down with uniform speed when its weight is balanced with the tension of its cable.

Statement II : Force exerted by the floor of an elevator on the foot of a person standing on it is more than his/her weight when the elevator goes down with increasing speed.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are false

(B) Both Statement I and Statement II are true

(C) Statement I is false but Statement II is true

(D) Statement I is true but Statement II is false

Q62. MCQ (Single Correct Answer)

Open solution

If $\overrightarrow{E}$ and $\overrightarrow{K}$ represent electric field and propagation vectors of the EM waves in vacuum, then magnetic field vector is given by :

(ω - angular frequency) :

(A) $\frac{1}{\omega}\left(\overrightarrow{K}\times\overrightarrow{E}\right)$

(B) $\overrightarrow{K}\times\overrightarrow{E}$

(C) $\omega\left(\overrightarrow{K}\times\overrightarrow{E}\right)$

(D) $\omega\left(\overrightarrow{E}\times\overrightarrow{K}\right)$

Q63. Numerical

Open solution

A hollow cylindrical conductor has length of 3.14 m, while its inner and outer diameters are 4 mm and 8 mm respectively. The resistance of the conductor is $n\times10^{-3}\Omega$. If the resistivity of the material is $2.4\times10^{-8}\Omega\mathrm{m}$, the value of n is _____.

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Q64. Numerical



Open solution

A block of mass 2 kg is attached with two identical springs of spring constant 20 N/m each. The block is placed on a frictionless surface and the ends of the springs are attached to rigid supports (see figure). When the mass is displaced from its equilibrium position, it executes a simple harmonic motion. The time period of oscillation is $\frac{\pi}{\sqrt{x}}$ in SI unit. The value of x is _____.

Q65. Numerical

Open solution

Vectors $a\hat{i}+b\hat{j}+\hat{k}$ and $2\hat{i}-3\hat{j}+4\hat{k}$ are perpendicular to each other when $3a+2b=7$, the ratio of a to b is $\frac{x}{2}$. The value of x is _____.

Q66. Numerical

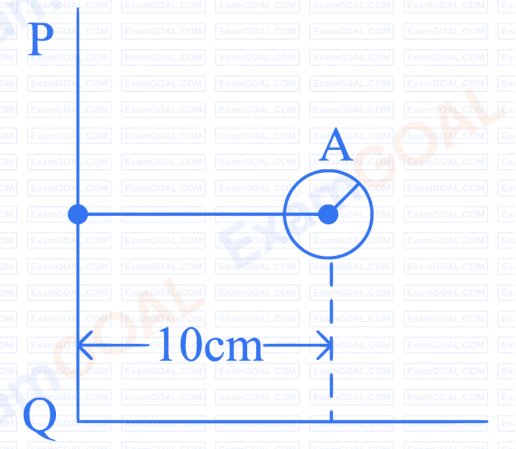
Open solution

A hole is drilled in a metal sheet. At $27^{\circ}\mathrm{C}$, the diameter of hole is 5 cm. When the sheet is heated to $177^{\circ}\mathrm{C}$, the change in the diameter of hole is $\mathrm{d}\times10^{-3}$ cm. The value of d will be _____ if coefficient of linear expansion of the metal is $1.6\times10^{-5}/^{\circ}\mathrm{C}$.

Q67. Numerical

Open solution

Solid sphere A is rotating about an axis PQ. If the radius of the sphere is 5 cm then its radius of gyration about PQ will be $\sqrt{x}$ cm. The value of x is _____.

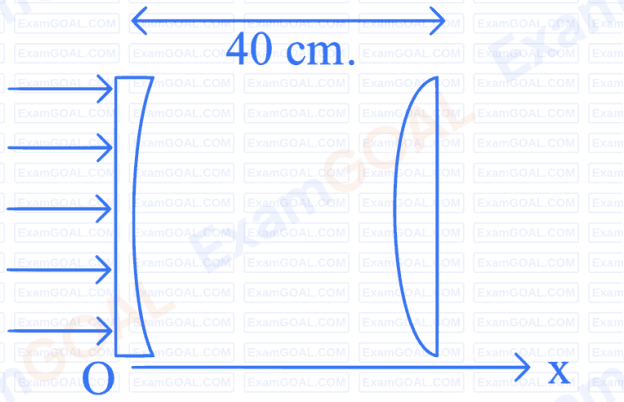


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Q68. Numerical

As shown in the figure, a combination of a thin plano concave lens and a thin plano convex lens is used to image an object placed at infinity. The radius of curvature of both the lenses is 30 cm and refraction index of the material for both the lenses is 1.75. Both the lenses are placed at distance of 40 cm from each other. Due to the combination, the image of the object is formed at distance $x =$ _____ cm, from concave lens.

Open solution



Q69. Numerical

A stream of positively charged particles having $\frac{q}{m} = 2 \times 10^{11} \frac{C}{kg}$ and velocity $\overrightarrow{v_0} = 3 \times 10^7 \hat{i}\, m/s$ is deflected by an electric field $1.8\hat{j}$ kV/m. The electric field exists in a region of 10 cm along x direction. Due to the electric field, the deflection of the charged particles in the y direction is _____ mm.

Open solution

Q70. Numerical

A spherical body of mass 2 kg starting from rest acquires a kinetic energy of 10000 J at the end of 5th second. The force acted on the body is _____ N.

Open solution

Out of syllabus 20 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q71. MCQ (Single Correct Answer) · **Chemistry**

Which of the following is true about freons?

(A) These are radicals of chlorine and chlorine monoxide

(B) These are chemicals causing skin cancer

(C) All radicals are called freons

(D) These are chlorofluorocarbon compounds

Open solution

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Q72. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Match List I with List II

	List-I		List-II
A.	Chlorophyll	I.	Na_2CO_3
B.	Soda ash	II.	$CaSO_4$
C.	Dentistry, Ornamental work	III.	Mg^{2+}
D.	Used in white washing	IV.	$Ca(OH)_2$

Choose the correct answer from the options given below :

(A) A-III, B-I, C-II, D-IV

(B) A-II, B-III, C-IV, D-I

(C) A-II, B-I, C-III, D-IV

(D) A-III, B-IV, C-I, D-II

Q73. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Statement I : For colloidal particles, the values of colligative properties are of small order as compared to values shown by true solutions at same concentration.

Statement II : For colloidal particles, the potential difference between the fixed layer and the diffused layer of same charges is called the electrokinetic potential or zeta potential.

In the light of the above statements, choose the correct answer from the options given below

(A) Both Statement I and Statement II are false

(B) Statement I is true and Statement II is false

(C) Both Statement I and Statement II are true

(D) Statement I is false but Statement II is true

Q74. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Match List I with List II

	List-I		List-II
A.	Reverberatory furnace	I.	Pig Iron
B.	Electrolytic cell	II.	Aluminium
C.	Blast furnace	III.	Silicon
D.	Zone Refining furnace	IV.	Copper

Choose the correct answer from the options given below :

(A) A-III, B-IV, C-I, D-II

(B) A-IV, B-II, C-I, D-III

(C) A-I, B-IV, C-II, D-III

(D) A-I, B-III, C-II, D-IV

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Q75. MCQ (Single Correct Answer) · **Chemistry**

Reaction of BeO with ammonia and hydrogen fluoride gives A which on thermal decomposition gives BeF_2 and NH_4F. What is 'A'?

(A) $(NH_4)Be_2F_5$ (B) $(NH_4)_2BeF_4$ (C) $(NH_4)BeF_3$ (D) H_3NBeF_3

Open solution

Q76. MCQ (Single Correct Answer) · **Chemistry**

Given below are two statements:

Statement I : Noradrenaline is a neurotransmitter.

Statement II : Low level of noradrenaline is not the cause of depression in human.

In the light of the above statements, choose the correct answer from the options given below

(A) Both Statement I and Statement II are incorrect

(B) Statement I is correct but Statement II is incorrect

(C) Both Statement I and Statement II are correct

(D) Statement I is incorrect but Statement II is correct

Open solution

Q77. MCQ (Single Correct Answer) · **Chemistry**

Which of the Phosphorus oxoacid can create silver mirror from $AgNO_3$ solution?

(A) $H_4P_2O_5$ (B) $H_4P_2O_7$ (C) $(HPO_3)_n$ (D) $H_4P_2O_6$

Open solution

Q78. MCQ (Single Correct Answer) · **Mathematics**

Let a tangent to the curve $y^2 = 24x$ meet the curve $xy = 2$ at the points A and B. Then the mid points of such line segments AB lie on a parabola with the :

(A) length of latus rectum 2 (B) directrix 4x = -3

(C) directrix 4x = 3 (D) length of latus rectum $\frac{3}{2}$

Open solution

Q79. MCQ (Single Correct Answer) · **Mathematics**

The value of $\sum_{r=0}^{22} {}^{22}C_r\,{}^{23}C_r$ is

(A) ${}^{44}C_{23}$ (B) ${}^{45}C_{23}$ (C) ${}^{44}C_{22}$ (D) ${}^{45}C_{24}$

Open solution

Q80. MCQ (Single Correct Answer) · **Mathematics**

The distance of the point (7, -3, -4) from the plane passing through the points (2, -3, 1), (-1, 1, -2) and (3, -4, 2) is :

(A) $4\sqrt{2}$ (B) 4 (C) 5 (D) $5\sqrt{2}$

Open solution

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Q81. MCQ (Single Correct Answer) · **Mathematics**

The compound statement $(\sim (P \wedge Q)) \vee ((\sim P) \wedge Q) \Rightarrow ((\sim P) \wedge (\sim Q))$ is equivalent to

(A) $((\sim P) \vee Q) \wedge (\sim Q)$

(B) $(\sim Q) \vee P$

(C) $((\sim P) \vee Q) \wedge ((\sim Q) \vee P)$

(D) $(\sim P) \vee Q$

Open solution

Q82. MCQ (Single Correct Answer) · **Mathematics**

The distance of the point $(-1, 9, -16)$ from the plane

$2x + 3y - z = 5$ measured parallel to the line

$\frac{x+4}{3} = \frac{2-y}{4} = \frac{z-3}{12}$ is :

(A) $13\sqrt{2}$

(B) 26

(C) $20\sqrt{2}$

(D) 31

Open solution

Q83. MCQ (Single Correct Answer) · **Mathematics**

Let $\vec{u} = \hat{i} - \hat{j} - 2\hat{k}, \vec{v} = 2\hat{i} + \hat{j} - \hat{k}, \vec{v}.\ \vec{w} = 2$ and $\vec{v} \times \vec{w} = \vec{u} + \lambda\vec{v}$. Then $\vec{u}.\ \vec{w}$ is equal to :

(A) $-\frac{2}{3}$

(B) $\frac{3}{2}$

(C) 2

(D) 1

Open solution

Q84. MCQ (Single Correct Answer) · **Mathematics**

Let α be a root of the equation $(a-c)x^2 + (b-a)x + (c-b) = 0$ where a, b, c are distinct real numbers such that the matrix $\begin{bmatrix} \alpha^2 & \alpha & 1 \\ 1 & 1 & 1 \\ a & b & c \end{bmatrix}$ is singular. Then, the value of $\frac{(a-c)^2}{(b-a)(c-b)} + \frac{(b-a)^2}{(a-c)(c-b)} + \frac{(c-b)^2}{(a-c)(b-a)}$ is

(A) 3

(B) 6

(C) 12

(D) 9

Open solution

Q85. Numerical · **Mathematics**

Let a tangent to the curve $9x^2 + 16y^2 = 144$ intersect the coordinate axes at the points A and B. Then, the minimum length of the line segment AB is _____

Open solution

Q86. Numerical · **Mathematics**

Suppose $\sum_{r=0}^{2023} r^2\ {}^{2023}C_r = 2023 \times \alpha \times 2^{2022}$. Then the value of α is _____

Open solution

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Q87. MCQ (Single Correct Answer) · **Physics**

A modulating signal is a square wave, as shown in the figure.



If the carrier wave is given as $c(t) = 2\sin(8\pi t)$ volts, the modulation index is :

(A) $\frac{1}{3}$ (B) 1 (C) $\frac{1}{4}$ (D) $\frac{1}{2}$

Open solution

Q88. MCQ (Single Correct Answer) · **Physics**

Consider the following radioactive decay process

$${}^{218}_{84}A \xrightarrow{\alpha} A_1 \xrightarrow{\beta^-} A_2 \xrightarrow{\gamma} A_3 \xrightarrow{\alpha} A_4 \xrightarrow{\beta^+} A_5 \xrightarrow{\gamma} A_6$$

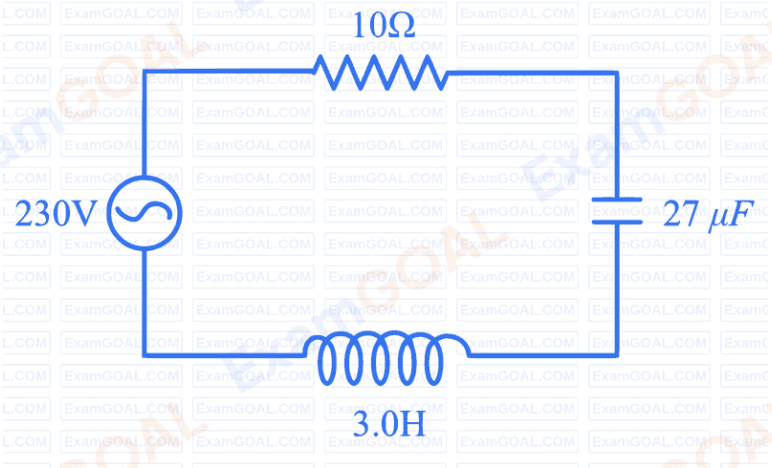
The mass number and the atomic number of A_6 are given by :

(A) 210 and 84 (B) 210 and 80 (C) 211 and 80 (D) 210 and 82

Open solution

Q89. Numerical · **Physics**

In the circuit shown in the figure, the ratio of the quality factor and the band width is _____ s.



Open solution

Q90. Numerical · **Physics**

Assume that protons and neutrons have equal masses. Mass of a nucleon is 1.6×10^{-27} kg and radius of nucleus is $1.5 \times 10^{-15}\ \mathrm{A}^{1/3}$ m. The approximate ratio of the nuclear density and water density is $n \times 10^{13}$. The value of n is _____.

Open solution

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Answer Key

Chemistry

1. **D**	2. **C**	3. **B**	4. **C**	5. **C**
6. **B**	7. **C**	8. **B**	9. **A**	10. **C**
11. **D**	12. **B**	13. **C**	14. **2**	15. **4**
16. **180**	17. **25**	18. **2**	19. **3**	20. **10**
21. **917**	22. **7**	23. **492**		

Mathematics

24. **B**	25. **C**	26. **D**	27. **B**	28. **B**
29. **C**	30. **A**	31. **B**	32. **C**	33. **C**
34. **D**	35. **D**	36. **A**	37. **5**	38. **14**
39. **12**	40. **118**	41. **546**	42. **22**	43. **2**
44. **60**				

Physics

45. **D**	46. **D**	47. **D**	48. **B**	49. **D**
50. **D**	51. **D**	52. **C**	53. **A**	54. **A**
55. **B**	56. **C**	57. **D**	58. **B**	59. **A**
60. **B**	61. **D**	62. **A**	63. **2**	64. **5**
65. **1**	66. **12**	67. **110**	68. **120**	69. **2**
70. **40**				

Out of syllabus

71. **D**	72. **A**	73. **B**	74. **B**	75. **B**
76. **B**	77. **A**	78. **C**	79. **B**	80. **D**
81. **C**	82. **B**	83. **D**	84. **A**	85. **7**
86. **1012**	87. **D**	88. **B**	89. **10**	90. **11**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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